

WASP-36b

R-0722

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_250405 · created 2026-07-28T00:03:46+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460770.7408 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.049
Transit depth (Rp/Rs)^2	2.77 +/- 1.62 [%]
Orbital Inclination (inc)	83.01 +/- 2.84
Airmass coefficient 1 (a1)	0.96 +/- 0.55
Airmass coefficient 2 (a2)	-0.09 +/- 0.17
Scatter in the residuals of the lightcurve fit is	49.81 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	5.5
Transit Duration (day)	0.071 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.049 vs 0.1368 (deviation 0.6σ)
Duration fit vs published	0.071 d vs 0.0758 d (deviation 0.19σ)
Mid-time O–C	-7.1 min
Depth SNR	3.4
Residual scatter	49.81 %

Light curve

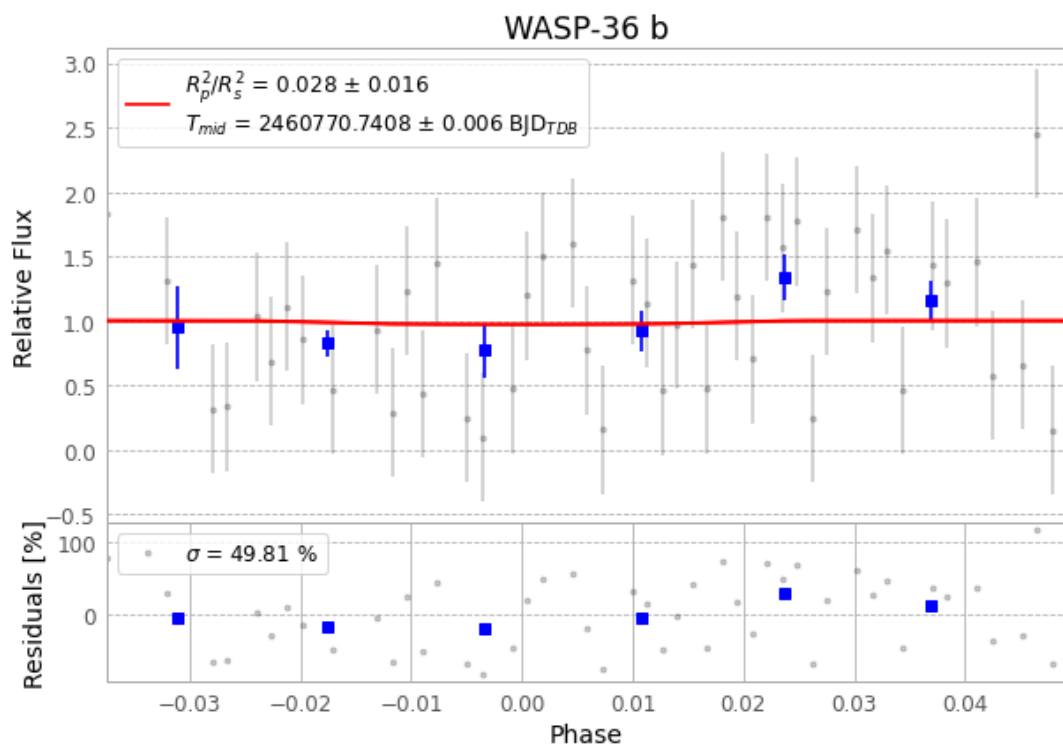


Figure 1 — FinalLightCurve_WASP-36 b_2025-04-04.png (SHA-256 6fef3f0be0fee4d3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [552, 446], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b3c605c41bcab617...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

69 FITS frames (45 MB), WASP-36250405040915.FITS → WASP-36250405073316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-250405.log (462c5787ff43...), FinalLightCurve_WASP-36 b_2025-04-04.png (6fef3f0be0fe...), FinalLightCurve_WASP-36 b_2025-04-04.csv (279b41f9ddd8...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 00:03Z.